

How roll, alloc and contiguous fill one sample

Data: ischemia_187_1_raw.mrd2, brightest of 312 series readouts (the averaged prescan is never shifted). 64 switches of 20 samples; the reconstruction reads positions 2..13 of each switch, with the echo expected at 8 (ramp 2 + 12/2).

Measured drift: +0.2021 samples per switch, so the gradient period is really 20.2021 where the stream is cut every 20. The echo walks +12.73 samples over the train. A +5 zero pad first puts switch 0's echo on position 8; the method then pulls every later switch back.

Switch i is displaced by $\text{offset}_i = -\text{slope} * i$ (anchored on switch 0). Output sample j of switch i is therefore read from the fractional position on the contiguous readout

$$\text{source} = i * \text{total} + j - \text{offset}_i = i * \text{total} + j + \text{slope} * i$$

which in general falls between two measured samples. The three methods differ only in which real samples they read around `source` and with what weight:

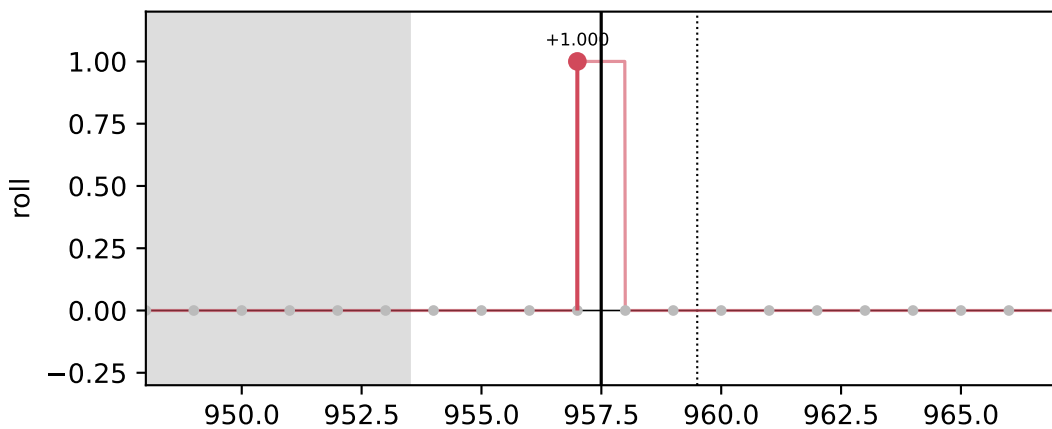
roll	$w = 1$ at $\text{round}(\text{source})$	1 tap
alloc	$w = 1-f$ at $\text{floor}(\text{source})$, f at $\text{floor}+1$	2 taps (linear)
contiguous	$w(x) = \text{sinc}(x) * \text{sinc}(x/8)$, $x = \text{source}-n$	16 taps (Lanczos-8)

f is the fractional part of source. The weights are real, so the same weights are applied to the real and to the imaginary part of the complex signal. All three read along the contiguous readout, so a tap past the end of a switch reads the real neighbouring switch rather than wrapping into its own ramp.

Worked switches: 47 ($f = 0.497$, near one half, where the kernels disagree most) and 26 ($f = 0.254$), both at output position $j = 8$. Every value in this document is checked against `mrd2shift.apply_roll / apply_alloc / apply_contiguous` when it is built.

Weight each method puts on the real samples around the fractional source (black line).
 Curve = continuous kernel, stems = the weights actually used; grey band = the source switch's kept window, dotted = switch boundaries

switch 47: source = 957.4970 (f = 0.497)



switch 26: source = 533.2537 (f = 0.254)

